Software Requirements Specification

Web-Based ML Model Evaluator

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the Web-Based ML Model Evaluator. It is the common reference for implementation and testing.

1.2 Scope

The system is a web application that allows users to upload tabular datasets, inspect and prepare the data, configure machine-learning experiments, train models, evaluate and compare results, and download supported outputs. Administrative and instructor functions are included only as described in the functional requirements.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
ML	Machine Learning
UI	User Interface
API	Application Programming Interface
CSV	Comma-Separated Values
Train-test split	Partitioning a dataset into training and testing subsets
Target column	Dataset column the model is intended to predict
Run	A configured model-training and evaluation execution

2. Overall Description

2.1 Product Perspective

The product is a browser-accessible application composed of a frontend, backend service, persistent storage, and an ML execution component. The frontend sends user actions to the backend. The backend validates requests, coordinates dataset and model operations, stores metadata and results, and returns responses for presentation.

2.2 Product Functions

- Register, authenticate, and manage a user session.
- Upload and validate CSV datasets; preview records and basic dataset statistics.
- Configure preprocessing, target column, train-test split, and supported ML algorithm.
- Train models and display applicable evaluation metrics and visualizations.
- Compare saved model runs, download supported artifacts/reports, and view run history.
- Provide administrative user/dataset/log functions and read-only instructor access to authorized evaluation results.

2.3 User Classes and Characteristics

Role	Description
Student / Registered User	Performs dataset upload, preparation, model configuration, training, evaluation, comparison, and downloads for authorized resources.
Instructor	Views evaluation results explicitly made available for review; access is read-only.
Admin	Manages user accounts, views/manages dataset records within the administrative interface, and views activity logs.
ML Service	Supporting training engine invoked by the application; it is a system dependency rather than a human user.

2.4 Operating Environment

The system is intended to run in a modern desktop browser such as Chrome, Firefox, or Edge. The backend, database, file/artifact storage, and ML libraries run in the configured project environment. Exact supported browser versions and deployment resources are to be recorded for the release.

3. Functional Requirements

The IDs below are stable references used by the Use Case Flow, Test Plan, and Security Requirements documents.

3.1 User Access and Session Management

ID	Requirement
FR-1.1	The system shall allow a user to register using the required account fields and validate those fields before creating an account.
FR-1.2	The system shall authenticate registered users using valid credentials and reject invalid credentials without creating an authenticated session.
FR-1.3	The system shall allow an authenticated user to log out and shall invalidate or terminate the active session.
FR-1.4	The system shall restrict protected application functions and user-specific resources to authenticated, authorized users.

3.2 Dataset Management

ID	Requirement
FR-2.1	The system shall allow an authenticated user to upload a CSV dataset.
FR-2.2	The system shall validate an uploaded dataset for supported file type, readable content, usable tabular structure, and required data presence before accepting it for evaluation.
FR-2.3	The system shall display a preview of the uploaded dataset, including column names and a bounded sample of rows.
FR-2.4	The system shall display basic dataset information, including row count, column count, detectable data types, and missing-value information where available.
FR-2.5	The system shall associate dataset metadata with its owner and allow the user to replace or upload another dataset without exposing another user's dataset.

3.3 Data Preparation

ID	Requirement
FR-3.1	The system shall provide supported preprocessing options for preparing a dataset for model training and validate preprocessing configuration before execution.
FR-3.2	The system shall allow the user to select a valid target column from the uploaded dataset.
FR-3.3	The system shall allow the user to configure the train-test split using supported proportions and validate that the resulting partitions are usable.

3.4 Model Configuration and Training

ID	Requirement
FR-4.1	The system shall display supported ML algorithms and allow the user to select an algorithm compatible with the configured task and dataset.
FR-4.2	The system shall train a model using the selected dataset, preprocessing configuration, target column, split, and algorithm.
FR-4.3	The system shall report training success or failure and associate each completed run with its configuration and results.

3.5 Evaluation, Comparison, and Export

ID	Requirement
FR-5.1	The system shall calculate and display applicable evaluation metrics for a trained model; classification and regression metrics shall be shown only when applicable to the task.
FR-5.2	The system shall display supported visualizations for model evaluation, such as a confusion matrix for classification or appropriate regression evaluation plots, where implemented.
FR-5.3	The system shall allow users to compare two or more available model runs using comparable metrics and configuration details.
FR-5.4	The system shall allow users to download available model artifacts and/or an evaluation report in formats supported by the implementation.

3.6 Run History and Service Interface

ID	Requirement
FR-6.1	The system shall retain a user-accessible history of prior model runs and their available status, configuration, metrics, and artifacts.
FR-6.2	The backend shall expose documented application endpoints for supported frontend operations and return structured success or error responses.
FR-6.3	The system shall display actionable validation or execution error messages when an upload, preprocessing operation, training run, or export cannot be completed.

3.7 Administration and Instructor Access

ID	Requirement
FR-7.1	The system shall allow an authenticated Admin to view registered user accounts and perform supported account-management actions.
FR-7.2	The system shall allow an authenticated Admin to view stored dataset records and metadata and perform supported dataset-management actions while preserving ownership and run associations.
FR-7.3	The system shall allow an authenticated Admin to view available application activity logs in read-only form.
FR-7.4	The system shall allow an Instructor to view evaluation results explicitly made available for review; this access shall be read-only.

4. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	For supported datasets within the configured size limit, common UI actions and metadata/preview requests shall return within an agreed target under normal load; the target shall be measured and documented during testing.
NFR-2	Performance	The system shall provide visible progress or a clear running status for model-training operations that take longer than a normal request.
NFR-3	Security	Passwords shall not be stored in plaintext; authentication credentials shall be handled using an appropriate secure password-storage mechanism.
NFR-4	Security	Server-side authorization shall protect user-specific datasets, model runs, and downloadable artifacts; hiding a UI control alone shall not count as access control.
NFR-5	Privacy	The system shall isolate each user's uploaded datasets, run history, metrics, and artifacts from other users unless explicit sharing is implemented.
NFR-6	Reliability	Invalid input or a failed training/export operation shall not corrupt previously saved datasets or completed run records.
NFR-7	Reproducibility	The system shall retain sufficient run configuration, including algorithm and relevant split/preprocessing settings, to explain how a result was produced.
NFR-8	Usability	The primary workflow shall be navigable through a desktop web UI with clear labels, validation messages, and status feedback.
NFR-9	Compatibility	The application shall support the agreed current versions of major desktop browsers; the exact test matrix shall be recorded for the release.
NFR-10	Maintainability	The implementation shall separate UI, API/business logic, data persistence, and model-training concerns into maintainable modules.
NFR-11	Scalability	The system shall enforce configurable upload/resource limits and handle datasets within the documented project capacity without uncontrolled resource use.
NFR-12	Data Integrity	Dataset metadata, run configuration, metrics, and artifact references shall remain consistently associated with the correct user and run.
NFR-13	Error Handling	Errors shown to users shall be understandable and shall not expose passwords, secret configuration, stack traces, or other sensitive internal details.
NFR-14	Portability	The application shall be deployable in the documented project environment using its setup instructions and declared dependencies.

5. Assumptions and Design Decisions

ID	Topic	Decision
A-1	Input format	CSV is the initial dataset input format unless another format is explicitly implemented.
A-2	Algorithms and metrics	Only algorithms and metrics implemented and documented by the project are supported.
A-3	Resource limits	Dataset size, upload limits, and training duration depend on configured server resources and must be documented.
A-4	User roles	Student/registered user, Instructor, and Admin

		are represented in the project documents; their functions must be present in the delivered scope or marked out of scope consistently.
A-5	Training data	Users are responsible for having permission to upload and process their datasets.
A-6	Task type	Metric availability depends on whether the configured task is classification or regression.
A-7	Preprocessing	The exact preprocessing operations and supported split proportions are limited to those implemented and documented.
A-8	Requirement numbering	Requirement IDs are stable; retired requirements shall not be reassigned to different behaviour.